

Minimal models from free fields

Simon Wood

The Australian National University

Joint work with
Olivier Blondeau-Fournier, Pierre Mathieu and David Ridout

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Outline

- ① Motivation
- ② Universal vs minimal vertex operator algebras
- ③ Free field realisations
- ④ Examples

How to build a conformal field theory

The two building blocks of a conformal field theory (CFT):

- A vertex operator algebra (VOA) V .
- A *physical* category $\mathcal{C}$ of V -modules which
 - is closed under conjugation
 $\Rightarrow$ non-degeneracy of 2pt functions,
 - is closed under fusion
 $\Rightarrow$ needed for 3pt functions,
 - admits a modular invariant partition function
 $\Rightarrow$ definable on torus.

Universal vs minimal vertex operator algebras

Universal vertex operator algebras form families of vertex operator algebras characterised by certain parameters (central charges, levels).

- simple for generic values of parameters – proper ideals trivial
- non-trivial proper ideals exist for special values of parameters
- **minimal models**: simple quotients at special values of parameters

Examples:

Universal Virasoro: parameter: central charge c ,

non-simple $\Leftrightarrow c = 1 - 6 \frac{(p-q)^2}{pq}$, $p, q \in \mathbb{Z}_{\geq 2}$, $\gcd\{p, q\} = 1$,
minimal model is unitary $\Leftrightarrow |p - q| = 1$.

Universal $\widehat{\mathfrak{sl}}(2)$: parameter: level k ,

non-simple $\Leftrightarrow k + 2 \in \mathbb{Q}^+ \setminus \{\frac{1}{m}\}_{m=1}^{\infty}$,
minimal model is unitary $\Leftrightarrow k \in \mathbb{Z}_{\geq 0}$,
minimal model is not unitary $\Leftrightarrow$ logarithmic

Universal vs minimal modules

- Module theory of universal vertex operator algebras very hard. Contains some easy families: Verma modules
- Universal VOA module M :
 M is a minimal model module $\Leftrightarrow$ Ideal in universal VOA acts trivially.
- Minimal model module theory strongly constrained

Example: Virasoro universal vs minimal modules

Universal: Any module with conformal weights bounded below, such as Verma modules, staggered modules and *many* more.

Minimal: $\frac{1}{2}(p-1)(q-1)$ simple modules, all modules semisimple.

Free field vertex operator algebras

Free field algebra: Operator product expansions of generating fields proportional to identity.

Canonical examples:

Neutral boson NB: $a(z)a(w) \sim \frac{1}{(z-w)^2}$

Neutral fermion NF: $\psi(z)\psi(w) \sim \frac{1}{z-w}$

Charged bosons CB: $\beta(z)\gamma(w) \sim \frac{-1}{z-w}, \quad \beta(z)\beta(w) \sim 0 \sim \gamma(z)\gamma(w)$

Charged fermions CF: $b(z)c(w) \sim \frac{1}{z-w}, \quad b(z)b(w) \sim 0 \sim c(z)c(w)$

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Free field realisations

- Universal vertex operator algebras $\xrightarrow{\text{embedding}}$ free field algebras.
 $\Rightarrow$ Free field modules also universal modules. [Feigin-Fuchs; Wakimoto; Kac-Frenkel]
- Free field module theory is easy.
- Ideals easy to construct in free field algebra.
 $\Rightarrow$ Can evaluate ideals on free field modules.

Examples:

- Virasoro $\hookrightarrow \text{NB}$
- $\widehat{\mathfrak{g}} \hookrightarrow \text{NB}^{\otimes \text{rk } \mathfrak{g}} \otimes \text{CB}^{\otimes \# \text{ pos. roots}}$
 $\Rightarrow \widehat{\mathfrak{sl}}(2) \hookrightarrow \text{NB} \otimes \text{CB}$
- $N = 1$ Virasoro $\hookrightarrow \text{NB} \otimes \text{NF}$
- $\widehat{\mathfrak{osp}}(1|2) \hookrightarrow \text{NB} \otimes \text{CB} \otimes \text{CF}$

Free field realisations

Universal vertex operator algebra V , free field algebra F :

$$V \hookrightarrow F = \mathbf{NB}^{\otimes r} \otimes \mathbf{CB}^{\otimes s} \otimes \mathbf{NF}^{\otimes t} \otimes \mathbf{CF}^{\otimes u}$$

- For every factor $\mathbf{NB}$ in $F \Rightarrow$ exists family of vertex operators $V_p(z), p \in \mathbb{C}$.
- Vertex operators define maps between free field modules
- Screening field: $S(z) = (\text{bosons and fermions}) V_p(z)$, such that operator product expansion of $S(z)$ with V is total derivative
- Screening operators: Residue of $S(z)$. Commutes with V .
 $\Rightarrow$ Residue of $S(z)$ defines V -module homomorphism.
[Dotsenko-Fateev; Tsuchiya-Kanie]
- Ideal of $V \subset F$ can be constructed as image of screening operators [Felder; Bernard-Felder; Bouwknegt-McCarthy-Pilch]

Examples of screening operators

- Virasoro $\hookrightarrow$ NB: $c = 1 - 6 \frac{(p-q)^2}{pq}$

$$S_+(z) = V_{\sqrt{\frac{2q}{p}}}(z), \quad S_-(z) = V_{-\sqrt{\frac{2p}{q}}}(z)$$

- $\widehat{\mathfrak{sl}}(2) \hookrightarrow \text{NB} \otimes \text{CB}$: $k = -2 + \frac{u}{v}$

$$S(z) = \beta(z) V_{-\sqrt{\frac{2v}{u}}}(z)$$

- $N=1$ Virasoro $\hookrightarrow \text{NB} \otimes \text{NF}$: $c = \frac{3}{2} - 3 \frac{(p-q)^2}{pq}$

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$$S(z) = (\beta(z)c(z) - b(z)) V_{-\sqrt{\frac{v}{u}}}(z)$$

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The appearance of symmetric polynomials

Ideal of V in F can be constructed as image of screening operators

$$S^{[r]} = \int S(z_1) \cdots S(z_r) dz_1 \cdots dz_r.$$

Product of screening fields expands as

$$\begin{aligned} S(z_1) \cdots S(z_r) &= (\text{bosons and fermions}) V_p(z_1) \cdots V_p(z_r) \\ &= \underbrace{(\text{bosons and fermions})}_{(\text{skew symmetric})} \underbrace{\prod_{1 \leq i < j \leq r} (z_i - z_j)^{p^2}}_{(\text{Vandermonde det})^{p^2}} \underbrace{V_p(z_1) \cdots V_p(z_r)}_{\text{symmetric}} : \cdot \end{aligned}$$

Vandermonde determinant defines inner product for symmetric polynomials

$$\langle f, g \rangle = \int \prod_{1 \leq i \neq j \leq r} (z_i - z_j)^{p^2/2} z_i^{(r-2)p^2/2} f(z^{-1}) g(z) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}.$$

Exists orthogonal basis called **Jack polynomials**.

Symmetric polynomials

Ring of symmetric polynomials $\Lambda_r = \mathbb{C}[z_1, \dots, z_r]^{\mathcal{S}_r}$.

Admits many bases such as *monomial symmetric polynomials*

$$m_{\lambda}(z) = \sum_{\sigma} z_1^{\sigma_1} \cdots z_r^{\sigma_r},$$

$\lambda = [\lambda_1, \dots, \lambda_r]$ partition, σ distinct permutations of λ .

Examples:

- $m_{[1]}(z_1, z_2) = m_{[1,0]}(z_1, z_2) = z_1 + z_2$
- $m_{[2,2]}(z_1, z_2) = z_1^2 z_2^2$
- $m_{[3,2,1]}(z_1, z_2, z_3) = z_1^3 z_2^2 z_3 + z_1^3 z_2 z_3^2 + z_1^2 z_2^3 z_3 + z_1^2 z_2 z_3^3 + z_1 z_2^3 z_3^2 + z_1 z_2^2 z_3^3$
- $m_{[3,2,1]}(z_1, z_2) = 0$

Jack symmetric polynomials

Jack symmetric polynomials $P_\lambda^t(z)$ are basis of Λ_r characterised by

① Upper triangular decomposition:

$$P_\lambda^t(z) = m_\lambda(z) + \sum_{\mu < \lambda} c(t; \lambda, \mu) m_\mu(z)$$

② Orthogonal:

$$\langle f, g \rangle_r^t = \int \prod_{1 \leq i \neq j \leq r} (z_i - z_j)^{1/t} \prod_{i=1}^r z_i^{(r-1)/t} f(z^{-1}) g(z) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

⇒ Screening operators distinguish Jack polynomials as **natural basis**.

Classifying simple minimal model modules

Evaluating ideals on free field modules $\Rightarrow$ Polynomial constraints on weights of (relaxed) highest weight vectors [Feigin-Ooguri-Nakanishi; Ridout-SW]

Virasoro: $c = 1 - 6 \frac{(p-q)^2}{pq}$

$$\left\langle \mathbf{P}_{[(q-1)^{p-1}]}^{p/q}, \prod_{i=1}^{p-1} (1 + z_i)^{\sqrt{\frac{2q}{p}} \tau} \right\rangle_{p-1}^{p/q} \propto \prod_{r,s} (h_\tau - h_{r,s}),$$

τ neutral boson weight; h_τ conformal weight; $r, s \in$ Kac table

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$$\widehat{\mathfrak{sl}}(2): k = -2 + \frac{u}{v}$$

$$\begin{aligned} \sum_{n=0}^{u-1} (-1)^n \binom{u-2-n+\sigma}{u-1-n} \sum_{m=0}^n \binom{u-1-m}{n-m} \\ \times \left\langle \mathbf{P}_{[v^m, (v-1)^{u-1-m}]}^{u/v}, \prod_{i=1}^{u-1} (1+z_i)^{-\sqrt{\frac{2v}{u}}\tau-1} \right\rangle_{u-1}^{u/v} \\ \propto g_u(\lambda_{\tau,\sigma}, h_{\tau}) \prod_{r,s} (h_{\tau} - h_{r,s}), \end{aligned}$$

τ neutral boson weight, σ charged boson weight; $\lambda_{\tau,\sigma}$ $\widehat{\mathfrak{sl}}(2)$ weight, h_{τ} conformal weight; $r, s \in$ Kac table

Classifying simple minimal model modules

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$$N = 1 \text{ Virasoro: } c = \frac{3}{2} - 3 \frac{(p-q)^2}{pq}$$

$$\text{NS: } \left\langle \mathbf{P}_{\delta}^{-3}, \prod_{i=1}^{p-1} (1+z_i) \sqrt{\frac{q}{p}} \tau \right\rangle_{p-1}^{2p/(q-p)} \propto \prod_{r,s} (h_{\tau} - h_{r,s}^{(\text{NS})})$$

$$\text{R: } \left\langle \sum_{\lambda} c_{\lambda} \mathbf{P}_{\lambda}^{-3}, \prod_{i=1}^{p-1} (1+z_i) \sqrt{\frac{q}{p}} \tau \right\rangle_{p-1}^{2p/(q-p)} \propto \prod_{r,s} (h_{\tau} - h_{r,s}^{(\text{R})})$$

τ neutral boson weight; h_{τ} conformal weight; $r, s \in \text{Kac table}$

Why P_δ^{-3} ?

P_δ^{-3} contribution due to fermions in screening operator $\psi(z)V_{\sqrt{\frac{q}{p}}}(z)$

$$\langle \psi(z_1) \cdots \psi(z_{2n}) \rangle = \frac{1}{z_1 - z_2} \frac{1}{z_3 - z_4} \cdots \frac{1}{z_{2n-1} - z_{2n}} + \text{skew symm}$$

$\Rightarrow \prod_{1 \leq i < j \leq 2n} (z_i - z_j) \langle \psi(z_1) \cdots \psi(z_{2n}) \rangle$ is

- symmetric
- vanishes if $z_1 = z_2 = z_3$

Theorem: [Feigin, Jimbo, Miwa, Mukhin]

Ideal of symmetric polynomials f , where $f = 0$ if $z_1 = z_2 = z_3$, is spanned by Jack polynomials P_λ^{-3} with $\lambda_i - \lambda_{i+2} \geq 2$.

Big picture and future plans

$\widehat{\mathfrak{sl}}(2)$		Virasoro
$\widehat{\mathfrak{sl}}(n)$		W_n
$\widehat{\mathfrak{osp}}(1 2)$		$N = 1$ Virasoro
$\widehat{\mathfrak{sl}}(2 1)$	$\xrightarrow{\text{Hamiltonian reduction}}$	$N = 2$ Virasoro
$\widehat{\mathfrak{osp}}(3 2)$		$N = 3$ Virasoro
$\widehat{\mathfrak{psl}}(2 2)$		
$\widehat{\mathfrak{d}}(2 1; \alpha)$		$N = 4$ Virasoro

Unitary cases well studied.

Goal: Study all minimal modules uniformly. Non-unitary cases may be logarithmic $\Rightarrow$ new examples of logarithmic theories.

Big picture and future plans

$$\widehat{\mathfrak{sl}}(2) \checkmark$$

Virasoro $\checkmark$

$$\widehat{\mathfrak{sl}}(n) ?$$

W_n ?

$$\widehat{\mathfrak{osp}}(1|2) (\checkmark)$$

$N = 1$ Virasoro $\checkmark$

$$\widehat{\mathfrak{sl}}(2|1) ?$$

Hamiltonian reduction $\rightarrow$

$N = 2$ Virasoro

$$\widehat{\mathfrak{osp}}(3|2)$$

$N = 3$ Virasoro

$$\widehat{\mathfrak{psl}}(2|2)$$

$N = 4$ Virasoro

$$\widehat{\mathfrak{d}}(2|1; \alpha)$$

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Thank you for coming!